PlainSpeak RAG

Jargon-to-Plain Language Translator

Powered by Retrieval-Augmented Generation + NLP

Hackathon Project Description | NLP Track | 2026

1. Project Overview

Project name	PlainSpeak RAG
Track	Natural Language Processing (NLP)
Category	Text Simplification + Retrieval AI
Team size	3–4 people
Duration	24–48 hours
Primary tech	RAG · spaCy · sentence-transformers · FAISS · T5 / Mistral-7B · Gradio/ Streamlit/ HuggingFace
Datasets	Newsela · Wikipedia Simple English · MedEASi · MultiLexNorm

Problem Statement

Every day, millions of people encounter text they cannot understand. Legal contracts, medical consent forms, government regulations, and financial disclosures are written at a reading level far above the average reader. A US medical consent form averages Grade 16 readability — equivalent to a PhD dissertation — yet it is handed to patients who may read at Grade 6 level.

Existing solutions rely on simple rule-based substitution or generic large language models that lack domain grounding. They frequently change meaning, miss technical nuance, or produce output that is still too complex for the target audience.

Our Solution

PlainSpeak RAG is a Retrieval-Augmented Generation pipeline that simplifies complex text to a user-chosen reading level while preserving meaning. Unlike a plain LLM approach, our system grounds every simplification in a curated knowledge base of real, human-verified plain-language examples retrieved at inference time. The result is measurably more accurate, more consistent, and more controllable than prompt-only approaches.

2. System Architecture

Pipeline Overview

The system processes user input through five sequential stages, each contributing a distinct NLP or retrieval function:

- **Stage 1 — Input.** The user submits complex text — a legal clause, medical form, or government document.
- **Stage 2 — NLP Preprocessing.** spaCy tokenizes the input, detects the domain (legal / medical / general) using keyword rules, and tags jargon terms using named entity recognition and custom terminology lists.
- **Stage 3 — RAG Retrieval.** Each sentence is embedded using the all-MiniLM-L6-v2 sentence-transformer model and queried against a FAISS vector index of 5,000+ complex-simple pairs, filtered by matching domain.
- **Stage 4 — FK Re-ranking.** Retrieved examples are re-ranked by their Flesch-Kincaid grade proximity to the user's target reading level. This re-ranking step is the core NLP contribution of the project.
- **Stage 5 — Grounded Generation.** A prompt is assembled: system instruction + 3 retrieved examples + reading level control token ([GRADE_5], [GRADE_8], or [EXPERT]) + input sentence. This is passed to T5-base or Mistral-7B-Instruct for final generation.

Key Technical Contributions

Domain-filtered retrieval

Rather than retrieving the globally nearest neighbors, the system filters the FAISS index by domain label before retrieval. This prevents a legal simplification example from being used to simplify a medical sentence, which would otherwise cause term substitution errors.

Reading level control token

A discrete control token prepended to each generation prompt steers the output toward a target Flesch-Kincaid grade. This gives users live, interactive control over output complexity — a feature not present in any of the baseline systems evaluated.

FK-based retrieval re-ranking

After initial vector retrieval, examples are re-scored by the absolute difference between their plain-version FK grade and the user's target grade. Examples closest to the target level are prioritized, improving output alignment with the chosen complexity setting.

3. Datasets

All datasets are freely available on HuggingFace Hub or via direct institutional download. No paid API or data licensing is required.

Dataset	Size	Domain	Use in project
Newsela	~130K pairs	General / News	Primary source of complex-simple aligned pairs across reading levels
Wikipedia Simple English	200K+ articles	General / Encyclopedia	Self-aligned pairs: complex Wikipedia ↔ Simple English version
MedEASi	~2,600 pairs	Medical	Medical text simplification with expert-verified plain-language rewrites
MultiLexNorm	~12K instances	Social / General	Word-level normalisation pairs; used for jargon term lookup tables

The knowledge base is built by sampling a balanced mix from the four datasets — approximately 5,000 complex-simple pairs total — then embedding the complex column and storing vectors alongside plain-version text and metadata (domain, source, FK grade of plain version) in a FAISS flat index.

4. Technology Stack

Layer	Tool / Library	Purpose
NLP preprocessing	spaCy 3.x	Tokenization, NER, domain detection, jargon tagging
Embeddings	sentence-transformers	all-MiniLM-L6-v2 for fast, high-quality sentence vectors
Vector database	FAISS / ChromaDB	Approximate nearest-neighbor search over the knowledge base
Orchestration	LangChain	RAG chain, prompt assembly, LLM interface
Generation model	T5-base / Mistral-7B	T5 for speed in demo; Mistral for quality if GPU available
Evaluation	textstat	Flesch-Kincaid grade computation on input and output
Evaluation	sacrebleu	BLEU score against retrieved reference for accuracy measure
Dataset access	HuggingFace datasets	Unified API for loading all four datasets
User interface	Gradio	Fast, interactive demo UI deployable in one command

5. Team Roles & Division of Work

Member	Role	Responsibilities
Person 1	NLP + RAG engineer	Preprocessing pipeline (spaCy), jargon extraction, sentence embeddings, FAISS index build, domain-filtered retrieval, FK re-ranking logic
Person 2	LLM + Prompt engineer	Prompt design, reading level control tokens, LangChain chain implementation, LLM API integration, output quality tuning and ablations
Person 3	Data + UI + Evaluation	Dataset download and indexing, Gradio UI, before/after FK scorer, BLEU evaluation, demo preparation, presentation slides

6. Evaluation Plan

The system is evaluated across four dimensions. All metrics are computed automatically and displayed live in the demo UI, allowing judges to observe real-time improvement.

Metric	Tool	What it measures
Flesch-Kincaid grade (before vs after)	textstat	Primary readability measure — target is a drop of 4+ grade levels for medical/legal input
BLEU score vs retrieved reference	sacrebleu	Lexical accuracy — how close the output is to a human-verified simplification of similar text
Semantic cosine similarity	sentence-transformers	Meaning preservation — ensures simplification did not alter the core meaning of the source
Retrieval precision@3	custom	RAG quality — verifies that the top-3 retrieved examples are domain-correct and level-appropriate

Baseline Comparison

We compare three configurations to demonstrate the contribution of each component:

- LLM only (no retrieval) — T5 / Mistral with simplification prompt but no retrieved examples.
- RAG without re-ranking — FAISS retrieval but top-k returned by vector distance only, no FK filtering.
- PlainSpeak RAG (full system) — domain-filtered retrieval + FK re-ranking + control token.

The FK grade drop and semantic similarity scores are expected to improve at each stage, providing a clear ablation story for the judges.

7. Demo Plan

Live Demo Flow (30 seconds)

- Open the Gradio UI — left panel shows complex text input, right panel shows plain output side by side.
- Paste a real section from a GDPR privacy policy (pre-loaded as a one-click example). FK grade is shown immediately: Grade 18.
- Set reading level slider to Grade 5. Hit Simplify. Output appears in under 3 seconds.
- FK grade after: Grade 5. The before/after score cards update in real time — this is the visual punchline.
- Show the Retrieved Examples panel below — 3 similar plain-language sentences pulled from the knowledge base, proving the RAG pipeline is active.
- Slide the reading level to Expert and re-run — output adjusts, FK grade rises to Grade 10. Demonstrates live level control.

Pre-loaded Demo Documents

- GDPR Article 6 — lawfulness of processing (legal domain)
- Medical consent form for a surgical procedure (medical domain)
- Egyptian government administrative regulation excerpt (government domain)

Fallback Strategy

If the LLM API is slow or unavailable during the demo, the system returns the top-1 retrieved example directly, bypassing generation. This still demonstrates the complete RAG retrieval pipeline and produces a valid simplification. The demo will never break.

8. Novelty & Differentiators

Most hackathon NLP projects build a generic chatbot or a plain LLM wrapper. PlainSpeak RAG is differentiated on five dimensions:

Dimension	Typical hackathon project	PlainSpeak RAG
Retrieval	No retrieval / basic keyword search	Dense vector retrieval over domain-filtered knowledge base
Level control	Fixed output style	User-controlled reading level with FK re-ranking
Evaluation	None or subjective	4 automatic metrics shown live in UI
Ablation study	Single configuration	3-step ablation: LLM-only → RAG → RAG+rerank
Social impact	Generic use case	Accessibility for patients, non-native speakers, citizens

9. Project System Prompt

The following is the recommended system prompt to use when creating this project in the Claude Projects panel. It encodes all context needed so the AI assistant is immediately helpful without wasting quota on re-explaining the project setup in every conversation.

```
--- COPY BELOW THIS LINE ---
You are a technical assistant for the PlainSpeak RAG hackathon project.

PROJECT SUMMARY
PlainSpeak RAG is a Retrieval-Augmented Generation pipeline that simplifies jargon-heavy text (legal, medical, government) to a user-chosen reading level (Grade 5 / Grade 8 / Expert) while preserving meaning. It uses domain-filtered FAISS retrieval, Flesch-Kincaid re-ranking, and a reading level control token to ground LLM generation in verified plain-language examples.

TECH STACK
- NLP: spaCy 3.x (tokenization, NER, jargon tagging, domain detection)
- Embeddings: sentence-transformers / all-MiniLM-L6-v2
- Vector DB: FAISS (primary) or ChromaDB
- Orchestration: LangChain
- LLM: T5-base (fast demo) or Mistral-7B-Instruct (quality)
- Evaluation: textstat (FK grade), sacrebleu (BLEU), cosine similarity
- UI: Gradio
- Datasets: Newsela, Wikipedia Simple English, MedEASi, MultiLexNorm (all on HuggingFace)

TEAM ROLES
- Person 1: NLP + RAG (spaCy pipeline, FAISS index, retrieval, re-ranking)
- Person 2: LLM + prompts (LangChain chain, control tokens, generation quality)
- Person 3: Data + UI + evaluation (dataset prep, Gradio UI, metrics, demo, slides)

KEY DESIGN DECISIONS
- Domain-filtered retrieval: filter FAISS index by domain label before ANN search
- FK re-ranking: after top-k retrieval, re-sort by |fk_plain - target_grade| ascending
- Control token: prepend [GRADE_5], [GRADE_8], or [EXPERT] to every prompt
- Fallback: if LLM fails, return top-1 retrieved example directly
- Evaluation shown live in UI: FK before/after, BLEU, cosine similarity

HACKATHON CONSTRAINTS
- 24-48 hour build window
- No paid APIs unless free tier is sufficient
- Must have a live, 30-second demo ready
- Knowledge base size: ~5,000 pairs (enough for demo, fast to index)

When asked for code, provide complete, runnable Python snippets. When asked for explanations, be concise and technical. Always flag if a design choice has a faster hackathon-friendly alternative.
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